Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/26/2008
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Project Title:	Sleep-Wake Medications in Space Flight: Review of Best Practices		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/01/2007	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-HPF00002		
Performance Goal No.:			
Performance Goal Text:			

This project developed an evidence-based summary for “best practices” regarding the use of pharmacological sleep-promoting agents and wake-promoting agents in space flight, to mitigate fatigue-related risks to performance. Evidence was integrated from the medical and scientific literature (via PubMed and Web of Science), from regulatory databases (FDA, DEA), and from available DOD sources, regarding the most effective and safe use of nine sleep-promoting agents,a benzodiazepine receptor antagonist,and three wake-promoting agents, to form a summary of “best practices” relevant to the physiological and operational conditions that occur in space. “Best practices” entails integration of the following factors in the evidence database to be developed: pharmacokinetics and pharmacodynamics, safety evidence, indication and efficacy evidence, abuse potential, evidence of interactions with other medications (especially those used in space), carry-over effects, and reversibility of drug effects and carryover effects in the event of an emergency. The project had three goals: (1) Make scientifically-based recommendations on which of the currently

Task Description:	available sleep-promoting medications and wake-promoting agents are most appropriate for use in space flight. (2) Identify unwanted consequences and ways to mitigate them for these pharmacological agents (e.g., carry-over sedation by sleep-promoting medications, and sleep-disruption by wake-promoting medications). (3) Ongoing surveillance of ground-based evidence on adverse events and undesirable consequences of sleep-promoting and wake-promoting medications being used in space. The “best practices” summary database linked elcetronically to scientific publications through the federal PubMed database. It was made available electronically to key NASA personnel in the Human Research Program of the Space and Life Sciences Directorate at Johnson Space Center. The importance of reducing fatigue-related performance errors in space by the optimal use of sleep-wake medications is very high, and the resulting “best practices” summary database will help Flight Surgeons, Astronauts, NASA BHP personnel, and NASA Management remain aware of the opportunities and risk mitigation needed in the pharmacological management of sleep-wake dynamics in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This electronic database is linked to the scientific literature on the most commonly used sleep-promoting and wake-promoting medications. It has signficiant benefits on Earth beyond its utility for space flight. Physicians in the military and civilian sectors responsible for the safe and effective use of sleep-wake medications can use the database to optimize their decision making regarding these medications.
Task Progress:	This project reviewed the scientific literature (e.g., PubMed and Web of Science), medical documents (e.g., package inserts, Physicians Desk Reference, FDA warnings), and authoritative reviews regarding the efficacy and safety of medications used in space flight to manage sleep-wake dynamics. Task progress was made on three goals. 1. Scientifically-based recommendations were developed for best practices regarding sleep-wake medication use in space flight. This was accomplished following interaction with JSC Flight Physicians, who are responsible for prescribing safe and effective sleep-wake medications used in space flight. It was determined that the most useful deliverable would be to have immediate electronic access to up-to-date medical, scientific and regulatory (FDA) information on the safety of the medications (at different dosages) most commonly used in space. In this way, the Flight Physicians can determine the “best practices” relative to what needs to be accomplished in spaceflight. 2. The unwanted consequences of sleep-wake medications and ways to mitigate them were identified. The database provides immediate access to up-to-date and comprehensive literature on the unwanted consequences (e.g., sedating side effects of hypnotic medications) for each of 9 of the most commonly used prescription hypnotics and 3 stimulants. There are links to evidence relevant to abuse potential, carry-over sedation by sleep-promoting medications, and sleep-disruption by wake-promoting medications. 3. The database also includes linkages to ground-based studies for adverse events. The database is designed to be added to regularly for all published data and information on adverse events associated with current sleep-promoting and wake-promoting medications used in space flight.
Bibliography Type:	Description: (Last Updated: 06/30/2026)